

MECHANICS OF SOLIDS

UNIT 1 – IMPORTANT QUESTIONS WITH ANSWERS

SHORT ANSWER QUESTIONS – 2 MARKS

1. What are temperature stresses? Write the expression for thermal stress in a constrained material.

Temperature stress is the stress developed in a material when its free expansion or contraction due to a change in temperature is prevented.

Thermal stress: $\sigma = E \alpha \Delta T$

where E is Young's modulus, α is the coefficient of thermal expansion and ΔT is the temperature change.

2. Define strength and ductility.

Strength: The ability of a material to withstand an applied load without failure.

Ductility: The ability of a material to undergo considerable plastic deformation before fracture.

3. Define the following terms: (i) Resilience (ii) Impact loading.

Resilience: The capacity of a material to absorb energy within its elastic limit and release it when the load is removed.

Impact loading: A load applied suddenly to a body, producing a high rate of deformation and stress.

4. Define a composite bar.

A composite bar is a bar made of two or more different materials rigidly connected together so that they act together under load.

5. Define strain energy and write the formula for strain energy stored in a material under tensile load.

Strain energy is the energy stored in a body due to deformation when an external load is applied.

$$SE = \frac{1}{2} P \delta L$$

6. Define stress and strain. Also, write the SI units of stress and strain.

Stress: Internal resisting force developed per unit cross-sectional area.

$$\sigma = P/A$$

SI unit of stress: Pa (N/m²).

Strain: Deformation per unit original dimension.

$$\epsilon = \delta L/L$$

Strain is dimensionless and has no SI unit.

7. Define proof resilience and modulus of resilience.

Proof resilience: The maximum strain energy stored in a body up to the elastic limit.

Modulus of resilience: Proof resilience per unit volume of the material.

Modulus of resilience = Proof resilience / Volume.

8. Define the following terms: (i) Thermal stress (ii) Thermal strain.

Thermal stress: Stress developed due to prevented thermal expansion or contraction.

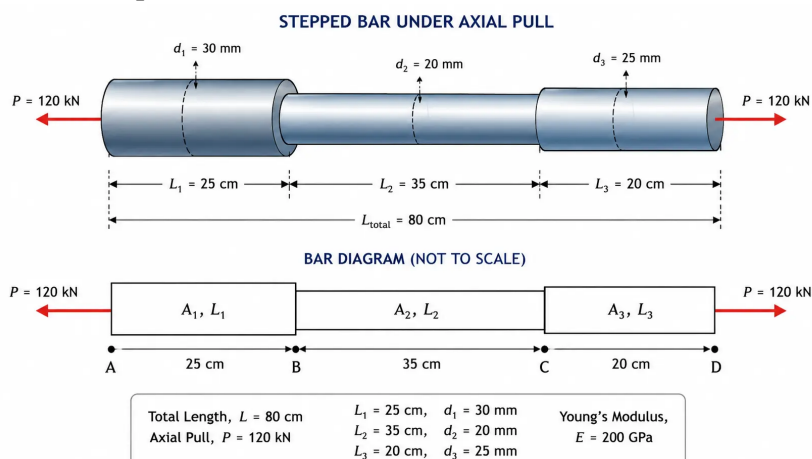
$$\sigma = E \alpha \Delta T$$

Thermal strain: Free strain produced due to a change in temperature.

$$\epsilon T = \alpha \Delta T$$

ESSAY / LONG ANSWER QUESTIONS

1. A straight bar 80 cm long consists of three portions. The first 25 cm length is of 30 mm diameter, the middle 35 cm length is of 20 mm diameter, and the remaining 20 cm length is of 25 mm diameter. If it is subjected to an axial pull of 120 kN, find the total extension of the bar. Find also the stresses, strains and changes in length of different portions. Take $E = 200,000 \text{ N/mm}^2$.



Given:

$$P = 120,000 \text{ N}$$

$$L_1 = 250 \text{ mm}, d_1 = 30 \text{ mm}$$

$$L_2 = 350 \text{ mm}, d_2 = 20 \text{ mm}$$

$$L_3 = 200 \text{ mm}, d_3 = 25 \text{ mm}$$

$$E = 200,000 \text{ N/mm}^2$$

Part I – Determination of stress, strain and extension:

$$A_1 = \pi/4 (30)^2 = 706.86 \text{ mm}^2$$

$$\sigma_1 = P/A_1$$

$$\sigma_1 = 120000/706.86 = 169.80 \text{ N/mm}^2$$

$$\epsilon_1 = \sigma_1/E$$

$$\epsilon_1 = 169.80/200000 = 8.490 \times 10^{-4}$$

$$\delta L_1 = PL_1/(EA_1)$$

$$\delta L_1 = [120000 \times 250]/[200000 \times 706.86]$$

$$\delta L_1 = 0.2122 \text{ mm}$$

Part II – Determination of stress, strain and extension:

$$A_2 = \pi/4 (20)^2 = 314.16 \text{ mm}^2$$

$$\sigma_2 = P/A_2$$

$$\sigma_2 = 120000/314.16 = 381.97 \text{ N/mm}^2$$

$$\epsilon_2 = \sigma_2/E$$

$$\epsilon_2 = 381.97/200000 = 1.910 \times 10^{-3}$$

$$\delta L_2 = PL_2/(EA_2)$$

$$\delta L_2 = [120000 \times 350]/[200000 \times 314.16]$$

$$\delta L_2 = 0.6685 \text{ mm}$$

Part III – Determination of stress, strain and extension:

$$A_3 = \pi/4 (25)^2 = 490.87 \text{ mm}^2$$

$$\sigma_3 = P/A_3$$

$$\sigma_3 = 120000/490.87 = 244.46 \text{ N/mm}^2$$

$$\epsilon_3 = \sigma_3/E$$

$$\epsilon_3 = 244.46/200000 = 1.222 \times 10^{-3}$$

$$\delta L_3 = PL_3/(EA_3)$$

$$\delta L_3 = [120000 \times 200]/[200000 \times 490.87]$$

$$\delta L_3 = 0.2445 \text{ mm}$$

Total extension:

$$\delta L = \delta L_1 + \delta L_2 + \delta L_3$$

$$\delta L = 0.2122 + 0.6685 + 0.2445$$

$$\delta L = 1.1252 \text{ mm}$$

2. A steel rod of 20 mm diameter and 200 mm length is subjected to a 20 kN tensile force. The extension is found to be 0.53 mm and the decrease in diameter is equal to 0.0022 mm. Determine: (i) Young's modulus of elasticity, (ii) Change in volume, and (iii) Poisson's ratio.

Given:

$$d = 20 \text{ mm}, L = 200 \text{ mm}$$

$$P = 20,000 \text{ N}$$

$$\delta L = 0.53 \text{ mm}$$

$$\delta d = 0.0022 \text{ mm}$$

(i) Young's modulus of elasticity:

$$A = \pi/4 (20)^2 = 314.16 \text{ mm}^2$$

$$E = PL/(A\delta L)$$

$$E = [20000 \times 200]/[314.16 \times 0.53]$$

$$E = 24,024.10 \text{ N/mm}^2$$

(ii) Change in volume:

$$\text{Initial volume, } V_i = \pi/4 d^2 L$$

$$V_i = \pi/4 (20)^2(200)$$

$$V_i = 62831.85 \text{ mm}^3$$

$$\text{Final length, } L_f = 200 + 0.53 = 200.53 \text{ mm}$$

$$\text{Final diameter, } d_f = 20 - 0.0022 = 19.9978 \text{ mm}$$

$$\text{Final volume, } V_f = \pi/4 (19.9978)^2(200.53)$$

$$V_f = 62984.50 \text{ mm}^3$$

$$\Delta V = V_f - V_i$$

$$\Delta V = 152.65 \text{ mm}^3$$

(iii) Poisson's ratio:

$$\text{Longitudinal strain, } \epsilon_{\text{long}} = 0.53/200 = 2.65 \times 10^{-3}$$

$$\text{Lateral strain, } \epsilon_{\text{lat}} = -0.0022/20 = -1.10 \times 10^{-4}$$

$$\mu = -\epsilon_{\text{lat}}/\epsilon_{\text{long}}$$

$$\mu = 0.0415$$

3. A specimen of steel 20 mm diameter with a gauge length of 200 mm is tested to destruction. It has an extension of 0.25 mm under a load of 80 kN and the load at elastic limit is 102 kN. The maximum load is 130 kN. The total extension at fracture is 56 mm and diameter at neck is 15 mm. Find: (i) Stress at elastic limit, (ii) Young's modulus, (iii) Percentage elongation, (iv) Percentage reduction in area, and (v) Ultimate tensile stress.

Given:

$$d = 20 \text{ mm, } L = 200 \text{ mm}$$

$$P = 80,000 \text{ N, } \delta L = 0.25 \text{ mm}$$

$$\text{Load at elastic limit, } P_{\text{elas}} = 102,000 \text{ N}$$

$$\text{Maximum load, } P_{\text{max}} = 130,000 \text{ N}$$

$$\text{Extension at fracture} = 56 \text{ mm}$$

$$\text{Diameter at neck} = 15 \text{ mm}$$

(i) Stress at elastic limit:

$$A_0 = \pi/4 (20)^2 = 314.16 \text{ mm}^2$$

$$\sigma_{\text{elastic}} = P_{\text{elas}}/A_0$$

$$\sigma_{\text{elastic}} = 102000/314.16$$

$$\sigma_{\text{elastic}} = 324.68 \text{ N/mm}^2$$

(ii) Young's modulus:

$$E = PL/(A_0 \delta L)$$

$$E = [80000 \times 200]/[314.16 \times 0.25]$$

$$E = 203,718.82 \text{ N/mm}^2$$

(iii) Percentage elongation:

$$L_f = 200 + 56 = 256 \text{ mm}$$

$$\% \text{ elongation} = [(L_f - L)/L] \times 100$$

$$\% \text{ elongation} = [(256 - 200)/200] \times 100$$

$$\% \text{ elongation} = 28\%$$

(iv) Percentage reduction in area:

$$A_f = \pi/4 (15)^2$$

$$\% \text{ reduction in area} = [(A_0 - A_f)/A_0] \times 100$$

$$\% \text{ reduction in area} = 43.75\%$$

(v) Ultimate tensile stress:

$$\sigma_{\text{ult}} = P_{\text{max}}/A_0$$

$$\sigma_{\text{ult}} = 130000/314.16$$

$$\sigma_{\text{ult}} = 413.80 \text{ N/mm}^2$$

4. The modulus of rigidity for a material is $0.5 \times 10^5 \text{ N/mm}^2$. A 10 mm diameter and 300 mm long rod of the material is subjected to an axial pull of 10 kN. The changes in diameter and length are 0.00015 mm and 0.015 mm respectively. Calculate Poisson's ratio and bulk modulus of elasticity.

Given:

$$G = 0.5 \times 10^5 \text{ N/mm}^2$$

$$d = 10 \text{ mm, } L = 300 \text{ mm}$$

$$P = 10,000 \text{ N}$$

$$\delta d = 0.00015 \text{ mm}, \delta L = 0.015 \text{ mm}$$

(i) Poisson's ratio:

$$\text{Lateral strain, } \epsilon_{\text{lat}} = \delta d/d$$

$$\epsilon_{\text{lat}} = 0.00015/10 = 0.000015$$

$$\text{Longitudinal strain, } \epsilon_{\text{long}} = \delta L/L$$

$$\epsilon_{\text{long}} = 0.015/300 = 0.00005$$

$$\mu = \epsilon_{\text{lat}}/\epsilon_{\text{long}}$$

$$\mu = 0.3$$

(ii) Young's modulus:

$$E = 2G(1 + \mu)$$

$$E = 2(0.5 \times 10^5)(1 + 0.3)$$

$$E = 1.3 \times 10^5 \text{ N/mm}^2$$

(iii) Bulk modulus:

$$K = E/[3(1 - 2\mu)]$$

$$K = 130000/[3(1 - 2 \times 0.3)]$$

$$K = 130000/1.2$$

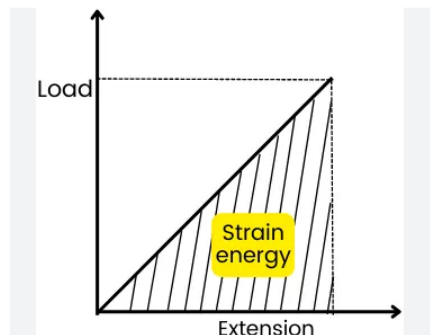
$$K = 108333.33 \text{ N/mm}^2$$

5. Define strain energy and derive the expression for strain energy stored in a material subjected to a tensile load. Also derive the expression for strain energy in terms of stress, Young's modulus and volume of the material.

Derivation:

Strain energy is the energy absorbed by the body during deformation.

In a load versus extension graph, the area under the line gives the strain energy.



$$SE = \text{Area of triangle}$$

$$SE = 1/2 \times P \times \delta L$$

$$\text{Also, } \sigma = P/A$$

$$\text{Therefore, } P = \sigma A$$

$$\text{Extension, } \delta L = PL/(EA)$$

$$\text{Substituting } P = \sigma A:$$

$$\delta L = \sigma L/E$$

$$\text{Therefore,}$$

$$SE = 1/2 \times \sigma A \times (\sigma L/E)$$

$$SE = \sigma^2 AL/(2E)$$

$$\text{Since } AL \text{ is the volume } V,$$

$$SE = \sigma^2 V/(2E)$$

Final expression:

$$SE = P^2 L/(2AE) = \sigma^2 V/(2E)$$